

DAKSH PANCHAL

+91 9910153794 dakshpanchal08@gmail.com LinkedIn GitHub

EDUCATION

Delhi Technological University **Aug 2024 – May 2028**
B.Tech Computer Science and Engineering — GPA: 8.875/10.00 (upto 3rd sem) *Delhi*
JEE Main 2024: AIR 11,214 (99.3 Percentile) **2024**

EXPERIENCE - RESEARCH TEAM : UNMANNED GROUND VEHICLE

Mars Exploration Drone — ISDC 2026 India | Repository | Report **2026**
– Engineered Pixhawk–Raspberry Pi flight stack with MAVLink, dual-pilot control, and manual compliance
– Integrated 64MP imaging, environmental sensing (SHT45, BME688, dust), and dual-RF telemetry

Autonomous Ground Vehicle — IGVC 2025 USA | Repository | Report **2025**
– a ROS-based autonomy stack with SLAM and DWA, integrating multi-sensor to achieve 95% waypoint accuracy

Team Achievements

10th Global Rank — ISDC 2026 | International Space Drone Challenge **Manipal, India — 2026**
Top 15 Global Rank — IGVC 2025 | Autonomous Navigation Challenge **Michigan, USA — 2025**

PERSONAL PROJECTS

Mars Exploration Drone Software Stack | GitHub **2026**
– Architected and implemented a Pixhawk–Raspberry Pi flight software stack using MAVLink, enabling dual-pilot control with <100ms latency, telemetry routing at 50Hz, and failsafe manual override compliance across 50+ flight test hours
– Built onboard compute pipeline integrating 64MP imaging and multi-sensor fusion (SHT45, BME688, particulate), with dual-RF telemetry for redundant communication

InLab - QR-Based Attendance System | GitHub **2026**
– Built a cross-platform attendance system using Flutter, Node.js, and PostgreSQL with GitHub OAuth authentication
– Implemented QR-based check-in/out with JWT-secured APIs, Dockerized backend, and real-time attendance tracking

POSITION OF RESPONSIBILITY

UGV-DTU Research Team **Aug 2024 – Present**
– **Software Head (2025–Present):** Lead a team of 15+ software engineers, managing software architecture, codebase development, and technical direction
– **Software Engineer (2024–2025):** Developed core autonomy and navigation modules

TECHNICAL SKILLS

Robotics: ROS2, SLAM, A*/DWA Planning, Kalman/EKF, PID/MPC Control, Gazebo, Sensor Fusion, LiDAR
Programming: C++17/20, Python (OpenCV, TensorFlow), Go, JavaScript/TypeScript
Systems & Embedded: Linux, Docker, Git, CI/CD, REST APIs, Microservices, Arduino, UART/SPI/I2C